

# Shivakumar Sridhar

shivakumrsridhar@gmail.com | +1 (602) 369-4928 | linkedin.com/in/shivakumar-s | github.com/shivakumrr

## Education

- **Arizona State University** Aug 2024 – April 2026  
*M.S. in Robotics and Autonomous Systems*  
Tempe, AZ
  - Coursework: Computer Vision, LLM, VLM, Diffusion GANs, Robot Kinematics, Motion Planning.
  - GPA: 3.63
- **Visvesvaraya Technological University** June 2016 – Aug 2020  
*B.E. in Computer Science and Engineering*  
Bengaluru, India
  - Coursework: Machine Learning, Data Structures, C++, Python, IoT, Computer Graphics.

## Experience

- **INDO-MIM Limited** May 2025 – Aug 2025  
*Computer Vision Engineer*  
San Antonio, TX
  - Built YOLO-driven perception, achieving 0.5 seconds inference for vision-driven high-speed robot inspections.
  - Optimized pytorch YOLO11 models, boosting defect detection accuracy from 60% to 99% and increasing throughput to 6,000 parts/day.
  - Architected full-stack MLOps pipelines from data collection to deployment via Acumen AI, productionized in 3 weeks.
  - Designed zero-shot LLM-vision pipeline enabling semantic multi-part picking without labeled training data.
- **Analytics Quad4** Jul 2022 – Dec 2023  
*Software Developer (IoT)*  
Bengaluru, India
  - Led a team of 4 to architect a B2B ML-SaaS, improving data ingestion efficiency by 60% via optimized pipelines.
  - Developed IoT edge pipelines for predictive maintenance, using LSTMs to forecast thermal anomalies.
  - Deployed A\* and Dijkstra's for real-time route optimization across 2,000+ vehicles, reducing logistics latency.
- **Entuple Technologies** Jun 2021 – Jul 2022  
*IoT Design and Support Engineer*  
Bengaluru, India
  - Lectured 400+ students on Supervised (SVM), Unsupervised (K-Means), and RL (DQN) for IoT apps.
  - Developed CNN-based Face ID for university attendance, deploying real-time vision software at scale.

## Projects

- **Denoising Diffusion GANS and Latent Space Analysis - Project Link** March 2025  
*Arizona State University*
  - Architected a DD-GAN pipeline in PyTorch, achieving a 600x sampling speed-up over standard DDPM models.
  - Executed distributed multi-GPU training on the SOL supercomputer, optimizing T=2 and T=4 diffusion steps.
  - Automated MLOps evaluation metrics, tracking FID (3.17) and Inception Scores (9.7) across 1,400 training epochs.
- **Cobot Path Planner - Project Link** Nov 2024  
*Arizona State University*
  - Developed an maze-solving pipeline for MyCobot Pro 600 using OpenCV for real-time topology extraction.
  - Architected CV-driven mapping to transform live video into discrete occupancy grids, enabling collision-free navigation.
  - Implemented Inverse Kinematics and A\* pathfinding in MATLAB/Simulink, achieving precise path execution.
- **Autonomous Drone Platform – Founder, CEO** Sep 2020 – May 2021  
*Bengaluru, India*
  - Developed a real-time SLAM and Visual Odometry pipeline for autonomous navigation using a ROS.
  - Implemented Extended Kalman Filters for sensor fusion, integrating IMU and GPS data to boost localization stability.
  - Recognized as one of the top 500 Innovative startups in Southeast Asia by Innopreneurs - [View Certificate](#).